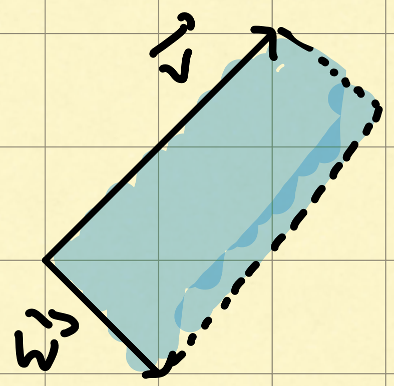
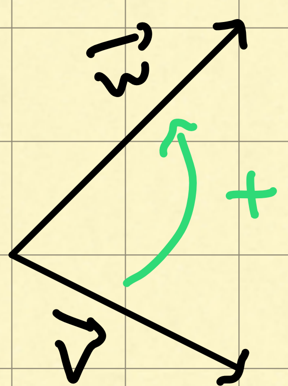
Cross Product

- cross product: standard intro

- 

consider parallelogram enclosed by $\vec{v}$ & $\vec{w}$

- $\vec{v} \times \vec{w}$ = area of this parellogram



if V to the right, $\vec{v} \times \vec{w}$ positive
if V left, $\vec{v} \times \vec{w}$, negative area

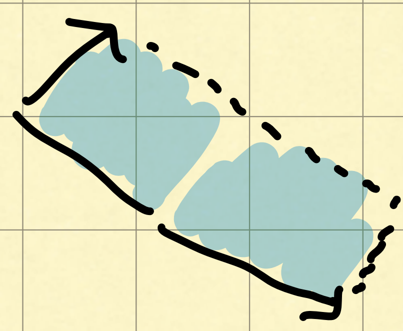
- order matters

- $\hat{i} \times \hat{j} = +1$. order of basis vectors is what defines orientation

- How to compute area? determinant
- $\underset{\vec{v}}{\begin{bmatrix}3\\1\end{bmatrix}} \times \underset{\vec{w}}{\begin{bmatrix}2\\-1\end{bmatrix}} = \det\left(\begin{bmatrix}3 & 2\\ 1 & -1\end{bmatrix}\right)$
- bc matrix whos cols represent v & w, represents lin transform that moves basis vectors to $\vec{v}$ & $\vec{w}$.

- det measures how area changes due to a transformation

- Perp. vectors → bigger cross product
- similar directions → smaller
- $(3\vec{v}) \times \vec{w} = 3(\vec{v} \times \vec{w})$



- true cross product combines 2 3D vectors to get a new 3D vector

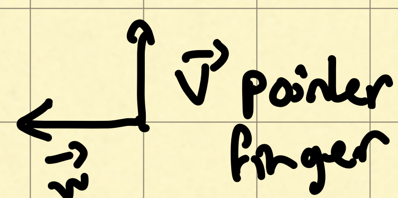
- still consider ▱ defined by 1st two vectors

- $\vec{v} \times \vec{w} = \vec{P}$, a vector
- $|\vec{P}|$ = area of parallelogram enclosed by $\vec{v}$ & $\vec{w}$.

- direction: perpendicular to the parallelogram

- but which way? u can be perpendicular in two directions

- right hand rule



thumb is direction of the cross product $\vec{v} \times \vec{w}$

- $\vec{v} \times \vec{w}$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \to \begin{bmatrix} \hat{\imath} & v_1 & w_1 \\ \hat{\jmath} & v_2 & w_2 \\ \hat{k} & v_3 & w_3 \end{bmatrix} \to \det\left(\begin{bmatrix} \dots \end{bmatrix}\right)$$

- but how to put vector as entry of a matrix?

- Students told that once u compute:

$$\hat{\imath}(\dots) + \hat{\jmath}(\dots) + \hat{k}(\dots)$$

is lin comb of basis vectors that represents $\vec{p}$, where:

$\to |\vec{p}| = \text{Area}_{\square}$ bw $\vec{v}$ & $\vec{w}$

$\to \measuredangle \vec{p} = \perp$ to $\square$ bw $\vec{v}$ & $\vec{w}$

- reason for this notational trick...